THINKORSWIM COMPLETE SETUP GUIDE - VXX TRADING SYSTEM 2026

With VXX/VIX Ratio Integration

OVERVIEW

This setup transforms ThinkOrSwim into a professional VXX trading machine with:

- **VXX/VIX Ratio** (the missing piece - value filter)
- **Volume confirmation** (institutional participation)
- **Moving averages** (trend and resistance)
- **VWAP** (institutional bias)
- **Alerts** (never miss a setup)

Setup Time: 15 minutes Maintenance: Zero (dynamic indicators, no daily updates)

STEP 1: BASIC CHART CONFIGURATION

Chart Settings:

1. Right-click chart → **Chart Settings**
2. **Time Frame**: 5 minutes (primary trading timeframe)
3. **Chart Type**: Candlestick
4. **Background**: Dark (easier for long sessions)
5. **Symbol**: VXX

STEP 2: ESSENTIAL INDICATORS (Add in Order)

A. 20-Period Simple Moving Average (Short-term Trend)

1. Click **Studies** → **Add Study** → Search: "SimpleMovingAvg"
2. Settings:
    - **Length**: 20
    - **Color**: Yellow

- **Line Weight**: 2
- **Label**: "20 SMA"

3. Click **OK**

Purpose: Short-term trend, immediate support/resistance

B. 50-Period Simple Moving Average (Medium-term Trend)

1. **Studies** → **Add Study** → Search: "SimpleMovingAvg"
2. Settings:
   - **Length**: 50
   - **Color**: Orange
   - **Line Weight**: 2
   - **Label**: "50 SMA"
3. Click **OK**

Purpose: Medium-term trend, major resistance level (VXX often fails here)

C. VWAP (Volume Weighted Average Price)

1. **Studies** → **Add Study** → Search: "VWAP"
2. Settings:
   - **Color**: Cyan/Light Blue
   - **Line Weight**: 2
   - **Time Period**: Day
3. Click **OK**

Purpose: Shows where institutions are positioned, acts as magnet and support/resistance

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D. Volume with Average (Confirmation Tool)

1. **Studies** → **Add Study** → Search: "Volume"
2. Settings:
   - **Show Volume Average**: ✅ CHECK THIS

- **Average Length**: 30
- **Above Average Color**: Green
- **Below Average Color**: Red
- **Volume Bars**: Yes

3. Click **OK**

Purpose: Volume >200% average = institutional participation = required for entry

Key Reading:

- Green bars = Above average (good)
- Red bars = Below average (skip)
- Current volume shows vs 30-period average in label

E. Implied Volatility (Options Environment)

1. **Studies → Add Study** → Search: "ImpVolatility"
2. Settings:
   - **Location**: Lower Panel (creates new subgraph)
   - **Color**: Cyan
   - **Show Current Level**: ✅ Check
3. Click **OK**

Purpose: Shows options pricing environment (avoid buying when IV >85%)

F. VXX/VIX RATIO ⭐ (THE MISSING PIECE)

This is your value filter - tells you when VXX is overpriced and worth fading

1. Click **Studies → Edit Studies → New**
2. Name it: "VXX_VIX_Ratio"
3. **Paste this exact code**:

```
thinkscript
```

```
declare lower;

def vix = close("VIX");
def vxx = close("VXX");
def ratioValue = vxx / vix;

plot RatioLine = ratioValue;
plot HighThreshold = 1.60;
plot LowThreshold = 1.35;

RatioLine.SetDefaultColor(Color.CYAN);
HighThreshold.SetDefaultColor(Color.GREEN);
LowThreshold.SetDefaultColor(Color.RED);

AddCloud(RatioLine, HighThreshold, Color.DARK_GREEN, Color.CURRENT);
```

4. Click **OK**
5. Apply to **Lower Panel**

Purpose:

- **>1.60** (above green line) = VXX expensive, PREMIUM FADE
- **1.45-1.55** (middle) = Normal fade territory
- **<1.35** (below red line) = VXX too cheap, DON'T FADE

This single number tells you if the fade is worth taking

Reading the Ratio:

- Normal range: 1.40-1.55
- Panic spikes: 1.60-1.80 (best fade opportunities)
- Post-spike lows: 1.30-1.40 (skip puts)
- Current average: ~1.46-1.48

STEP 3: CRITICAL ALERTS SETUP

A. VXX Percentage Alerts

Alert 1: 3% Spike

1. Right-click chart → **Create Alert**
2. Settings:
    - Symbol: VXX
    - Condition: "% Change" "is greater than" "3.0"
    - Message: "VXX spike 3%+ - fade opportunity"
    - Sound: ✅ Enable
    - Frequency: Once per bar

Alert 2: 5% Major Spike

1. Condition: "% Change" "is greater than" "5.0"
2. Message: "VXX major spike 5%+ - PREMIUM FADE"

Purpose: Catches volatility spikes automatically

B. Volume Surge Alert

1. Right-click chart → **Create Alert**
2. Condition: "Volume" "is greater than" "Average(Volume,20) * 1.5"
3. Message: "VXX volume surge 150%+"
4. Sound: ✅ Enable

Purpose: Flags institutional participation

C. Time Window Alerts (Critical)

Alert 1: Morning Window

- Time: 9:45 AM
- Message: "Morning window in 5 minutes (9:50-10:15)"

Alert 2: Power Hour Window

- Time: 3:05 PM
- Message: "Power hour window in 5 minutes (3:10-3:25)"

Alert 3: Institutional Flow Window

- Time: 3:40 PM
- Message: "INSTITUTIONAL WINDOW in 5 min (3:45-4:10) - SUPREME SETUP"

Purpose: Never miss your highest probability trading windows

D. VIX Regime Alerts

Alert 1: VIX Warning

1. Symbol: VIX
2. Condition: "Last" "crosses above" "25"
3. Message: "VIX above 25 - REDUCE SIZE or PAUSE"

Alert 2: VIX Extreme

1. Condition: "Last" "crosses above" "30"
2. Message: "VIX above 30 - CRISIS MODE - CLOSE POSITIONS"

Purpose: Protects you during regime changes

STEP 4: WORKSPACE OPTIMIZATION

Save Your Setup

1. **Setup → Save Workspace As**
2. Name: "VXX_Trading_System_2026"
3. Include: ✅ All settings and alerts

Now you can load everything with one click

Optional: Multi-Timeframe Grid

If you want multiple views:

1. Click grid icon (top right)
2. Select **2x2** or **1x3** layout
3. Setup:
    - **Panel 1**: VXX 5-minute (your primary trading chart)

- **Panel 2**: VXX 1-minute (precision entries)
- **Panel 3**: VIX 5-minute (context for regime)
- **Panel 4**: Options chain

STEP 5: OPTIONS CHAIN SETUP

Quick Access Configuration

1. **Trade** tab → **Options**
2. Symbol: VXX
3. Settings:
   - **Expiration**: Current week (0-4 DTE)
   - **View**: Puts (your primary strategy)
   - **Strikes**: Show ±5 from current price
   - **Layout**: Single view

This gives you one-click access to option pricing

COMPLETE INDICATOR REFERENCE

Your Chart Should Show (Top to Bottom):

Main Chart Panel:

- Candlesticks (5-minute)
- Yellow line (20 SMA) - short-term trend
- Orange line (50 SMA) - resistance zone
- Cyan line (VWAP) - institutional bias

Volume Panel (below main chart):

- Blue bars (volume)
- Green/Red bars (above/below 30-period average)
- Label shows: Current volume vs Average volume

Implied Volatility Panel:

- Cyan line showing IV%
- Watch for IV >80% (expensive options)

VXX/VIX Ratio Panel ⭐ :

- Cyan line (current ratio)
- Green line at 1.60 (premium fade zone)
- Red line at 1.35 (no fade zone)
- Dark green cloud when ratio >1.60

DAILY USAGE WORKFLOW

Morning (8:30 AM):

1. Open ThinkOrSwim
2. Load "VXX_Trading_System_2026" workspace
3. Check overnight alerts
4. Verify all indicators loading

Pre-Market Check:

- VXX/VIX Ratio: ____
- Current VXX price: ____
- VIX level: ____
- Any overnight news: ____

During Trading:

- **Let alerts notify you** (don't stare at screen)
- **Check ratio before every trade**
- **Verify volume >200% average**
- **Exit by 3:55 PM** (4:05 PM if institutional flow)

Evening:

- Review day's trades
- Check if any alerts need adjustment (rare)

- Platform auto-saves your settings

READING YOUR CHART - QUICK GUIDE

Perfect Fade Setup Looks Like:

Main Chart:

- VXX price above 50 SMA (orange) or at resistance
- Shooting star or doji candlestick forming
- Price rejecting VWAP or moving average

Volume Panel:

- GREEN bars (above average)
- Volume showing >200% of 30-period average

VXX/VIX Ratio Panel:

- Cyan line above 1.50 (ideally >1.60)
- Approaching or above green threshold line

Time:

- 9:50-10:15 AM, 3:10-3:25 PM, or 3:45-4:10 PM

= GO FOR PUTS

Skip Trade When:

Volume Panel:

- RED bars (below average)
- Volume <200% average

VXX/VIX Ratio Panel:

- Cyan line <1.45
- Approaching red threshold line

Main Chart:

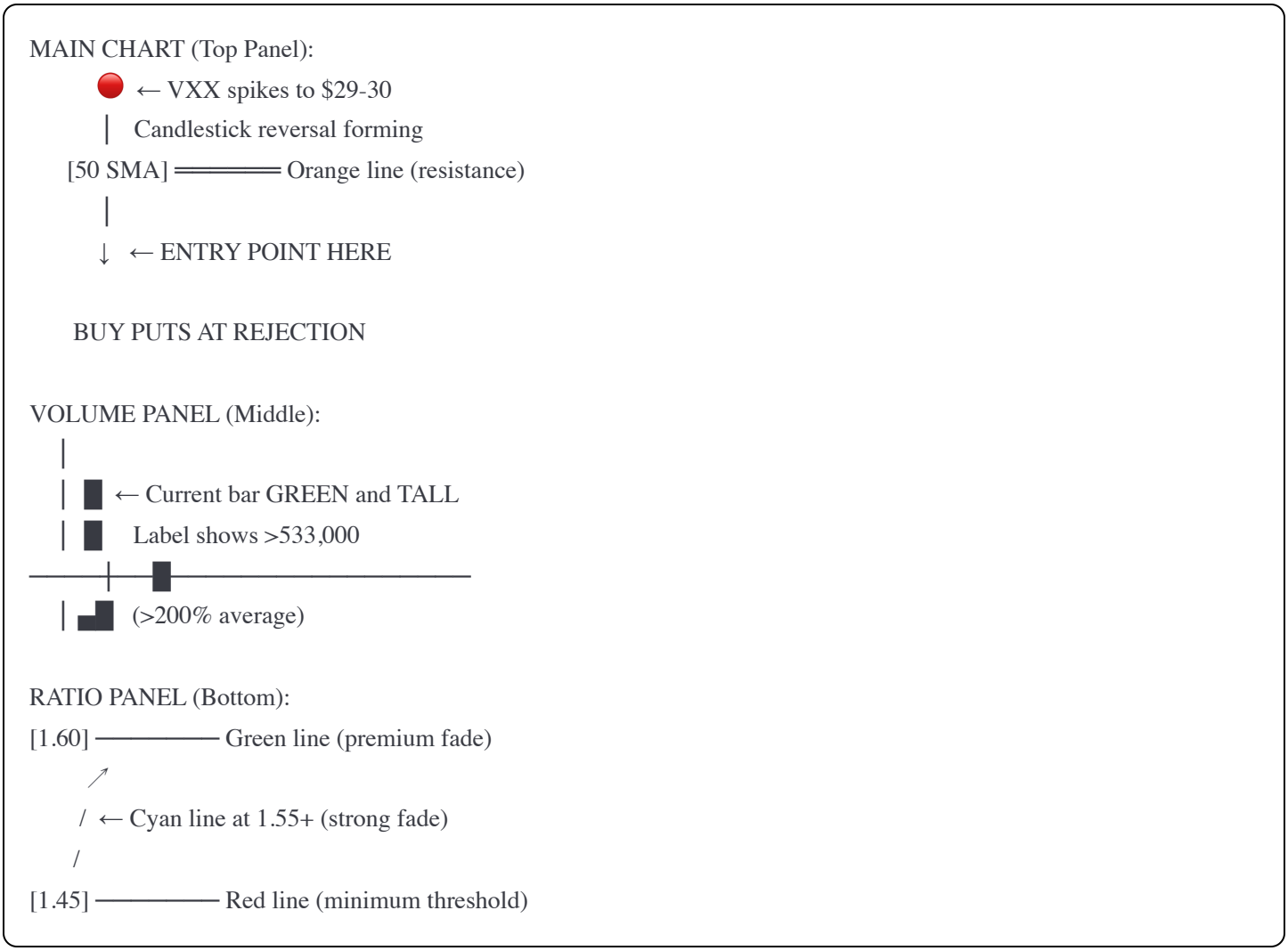
- No clear pattern
- Choppy, consolidating price action

= NO TRADE

VISUAL ENTRY & EXIT GUIDE

THE PERFECT ENTRY SETUP (What to Watch For)

At 3:10 PM, Your Chart Should Show:



This visual = IMMEDIATE ENTRY

CANDLESTICK PATTERNS TO WATCH

Best Reversal Signals for Puts:

1. Shooting Star ⭐ (Best signal):

```
 |  ← Long upper wick (buyers rejected)
 ■  ← Small red/green body
    ← Little or no lower wick
```

Meaning: Price tried to go higher, sellers crushed it **Action**: Enter puts immediately

2. Doji (Indecision at key level):

```
 |  ← Upper wick
 -  ← Tiny body (open ≈ close)
 |  ← Lower wick
```

Meaning: Battle at resistance, usually breaks down **Action**: Enter puts on next candle confirmation

3. Bearish Engulfing (Strong reversal):

```
[Previous]  [Current]
   □        ■■■  ← Large red candle
   □             ← Completely covers prior green
```

Meaning: Sellers took full control **Action**: Enter puts aggressively

4. Hanging Man (Top reversal):

```
    ← Little upper wick
 ■  ← Small body at top
 |  ← Long lower wick
```

Meaning: Failed rally attempt **Action**: Enter puts if at resistance

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THE 60-SECOND ENTRY PROCESS

At 3:10 PM Window Open:

Seconds 0-10: Quick Visual Scan

Main Chart Check:
├─ VXX location? (At $29-30 resistance?)
├─ At 50 SMA orange line? (YES/NO)
└─ Pattern forming? (Shooting star visible?)

Decision: Continue or Skip

Seconds 10-20: Volume Confirmation

Volume Panel Check:
├─ Current bar color? (GREEN/RED)
├─ Bar height? (Taller than average?)
└─ Label reading? (>533,000 = YES)

Decision: Continue or Skip

Seconds 20-30: Ratio Verification

Ratio Panel Check:
├─ Cyan line position? (Read number)
├─ Above 1.45? (Minimum required)
└─ Above 1.55? (Premium setup)

Decision: Position size determined

Seconds 30-40: Pattern Confirmation

Wait for Candle:
├─ Watch current 5-min candle complete
├─ Wick forming at top? (Rejection)
└─ Body closing lower? (Sellers won)

Decision: GO or Wait 1 more candle

Seconds 40-60: Execute Entry

Options Chain:
├─ Select strike ($29 puts if VXX at $29.50)
├─ Check bid/ask spread
├─ Enter limit order at mid-price
└─ Submit order

Decision: DONE - Position entered

Total Time: 60 seconds from setup to filled

RATIO + PATTERN POWER COMBINATION

Understanding the Synergy:

Just Pattern (No Ratio Check):

- Shooting star at $29 = 60% probability
- Could work, could fail
- Missing value filter

Just Ratio (No Pattern):

- Ratio 1.58 (good value) = 60% probability
- But where's the entry?
- Missing technical trigger

Pattern + Ratio Together:

- Shooting star at $29 + Ratio 1.58 = 85% probability ✅
- Value confirmed + Technical entry
- This is your edge

Both together validate each other:

- Ratio says "VXX is overpriced" (WHEN to trade)
- Pattern says "VXX is rejecting here" (WHERE to enter)

REAL-TIME RATIO EVOLUTION EXAMPLE

How Ratio Changes During Spike:

9:00 AM - Market Open:

- VXX: $28.09, VIX: 19.22
- Ratio: 1.46 (normal)
- Assessment: Standard fade zone

12:30 PM - Lunch:

- VXX: $28.29, VIX: 19.48
- Ratio: 1.45 (minimum threshold)
- Assessment: Borderline, would need confirmation

3:10 PM - Early Power Hour:

- VXX: $29.50, VIX: 19.80
- Ratio: 1.49 (rising)
- Assessment: Standard fade setup

3:15 PM - Spike Accelerates:

- VXX: $30.50, VIX: 19.50
- Ratio: 1.56 (strong)
- Assessment: Premium fade, full position ⭐

Key Insight: Ratio climbs when VXX spikes faster than VIX rises = overpriced = fade opportunity

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THE 3:10 PM ENTRY CHECKLIST

Print This - Keep Visible:

```
┌─────────────────────────────────────────────────────┐
│   3:10 PM POWER HOUR ENTRY CHECKLIST    │
├─────────────────────────────────────────────────────┤
│                                 │
│ [ ] TIME: 3:10-3:25 PM window active    │
│                                 │
│ [ ] PRICE: VXX at resistance ($29-30+)  │
│    OR breaking key support          │
│                                 │
│ [ ] PATTERN: Shooting star, doji, or    │
│    rejection candle visible         │
│                                 │
│ [ ] VOLUME: >533,000 (>200% average)    │
│    Green bars showing               │
│                                 │
│ [ ] RATIO: Cyan line >1.45          │
│    (Ideally >1.55 for full size)    │
│                                 │
│ ──────────────────────────────────────  │
│                                 │
│ ALL 5 CHECKED = ENTER IMMEDIATELY       │
│ 4 OR FEWER = SKIP OR WAIT           │
│                                 │
└─────────────────────────────────────────────────────┘
```

POSITION SIZING BY RATIO (At Entry)

Base your position size on ratio reading:

RATIO >1.60 (Above green line):
├─ Status: PREMIUM FADE
├─ Position: $500 (maximum)
├─ Contracts: 5-8 (depending on option price)
└─ Confidence: Highest

RATIO 1.55-1.60 (Approaching green):
├─ Status: STRONG FADE
├─ Position: $400-500
├─ Contracts: 4-6
└─ Confidence: High

RATIO 1.45-1.55 (Between lines):
├─ Status: NORMAL FADE
├─ Position: $300-400
├─ Contracts: 3-5
└─ Confidence: Standard

RATIO 1.35-1.45 (Approaching red):
├─ Status: WEAK FADE
├─ Position: $150-250 or SKIP
├─ Contracts: 1-2
└─ Confidence: Low

RATIO <1.35 (Below red line):
├─ Status: NO FADE
├─ Position: $0 (skip entirely)
├─ Contracts: 0
└─ Action: Wait for better setup

Never override ratio-based sizing, even when "certain"

EXIT STRATEGY - EQUALLY IMPORTANT

Three Exit Types (Use Whichever Hits First):

1. Profit Target Exit (Best Case):

Entry: $1.00 per contract
Target 1 (30%): $1.30 - Close 50% position
Target 2 (50%): $1.50 - Close remaining 50%
Result: Lock in gains, remove risk

2. Stop Loss Exit (Protection):

Entry: $1.00 per contract
Stop (50% loss): $0.50 - Close entire position
Result: Controlled loss, preserved capital
No hope, no waiting, mechanical exit

3. Time Stop Exit (Mandatory):

Entry: Any time before 3:25 PM
Exit: 3:55 PM SHARP (no exceptions)
Result: No overnight risk, system integrity

Exception: Institutional flow window
├─ Entry: 3:45-4:00 PM
├─ Exit: 4:05 PM maximum
└─ Still no overnight holds

Critical: Time stop overrides everything

- Even if position is winning
- Even if "just 5 more minutes"
- Even if "gap down likely overnight"

3:55 PM = exit. Period.

WHAT YOUR CHART SHOWS AT EXIT SIGNALS

Profit Target Hit (Exit Winner):

MAIN CHART:
VXX dropped from $29.50 → $28.00
Put option: $1.00 → $1.50 (50% gain)

ACTION:
├─ Close entire position
├─ Take profit
└─ Done for the day

Stop Loss Hit (Exit Loser):

MAIN CHART:
VXX rallied from $29.50 → $31.00
Put option: $1.00 → $0.50 (50% loss)

ACTION:
├─ Close entire position immediately
├─ Accept loss
├─ No revenge trading
└─ Review what criteria failed

Time Stop (3:55 PM Exit):

CLOCK: 3:55 PM

REGARDLESS OF P&L:
├─ Winning 20%? Close it.
├─ Losing 20%? Close it.
├─ Breaking even? Close it.
└─ "Just wants to see 4pm"? TOO BAD. Close it.

NO OVERNIGHT HOLDS = NON-NEGOTIABLE

POST-ENTRY MONITORING (The Watch)

What to Watch While In Position:

First 5 Minutes (3:10-3:15 PM):

Is trade working immediately?
├─ VXX dropping? Good, thesis working
├─ VXX rallying? Watch for stop
└─ VXX sideways? Normal, give it time

Next 15 Minutes (3:15-3:30 PM):

Check profit target:
├─ Up 30%? Consider scaling out 50%
├─ Up 50%? Close entire position
└─ Flat/down? Monitor stop level

Final Period (3:30-3:55 PM):

Approach time exit:
├─ 3:50 PM: Prepare to close
├─ 3:54 PM: Start exit process
└─ 3:55 PM: Position MUST be closed

Don't stare at every tick. Check every 5 minutes.

COMMON ENTRY MISTAKES TO AVOID

Mistake 1: "Close Enough" Syndrome

❌ "Ratio is 1.44, close to 1.45"
✅ NO. Hard rule is >1.45

❌ "Volume is 190%, almost 200%"
✅ NO. Must be >200%

❌ "It's 3:08, basically 3:10"
✅ NO. Wait for window

Mistake 2: Partial Criteria

❌ "Great pattern but low volume, I'll trade anyway"
✅ NO. ALL 5 criteria required

❌ "Perfect ratio but no pattern yet, I'll enter now"
✅ NO. Wait for pattern confirmation

Mistake 3: Prediction Override

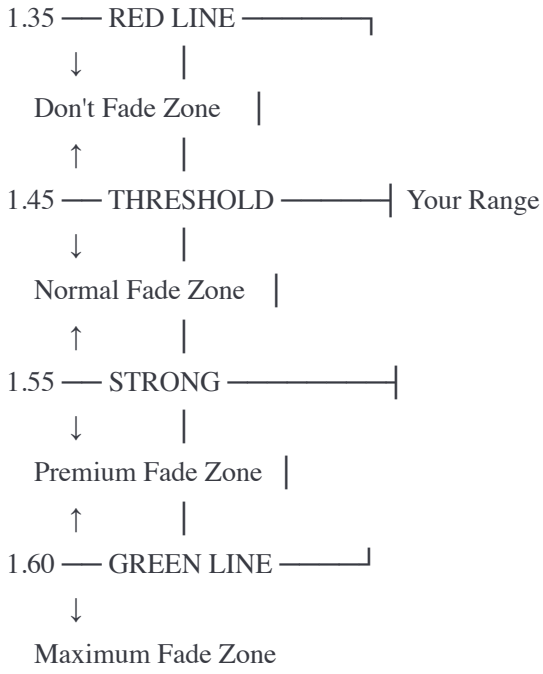
❌ "I know it's going down, criteria don't matter"
✅ NO. System over feelings

❌ "News says it should spike, I'll wait"
✅ NO. Trade setup, not predictions

The criteria exist to protect you. Honor them.

QUICK RATIO MEMORY GUIDE

Four Numbers to Memorize:



Current ratio: 1.45 = Right at minimum Watch for climb to 1.50-1.60 in power hour

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INDICATOR HIERARCHY & SYSTEM LOGIC

Understanding WHY Each Indicator Matters

This system uses 5 indicators. **All 5 are required**, but understanding their purpose and weight helps you execute with confidence.

THE WEIGHT & PURPOSE BREAKDOWN

Think of this like flying a plane - you need all systems working, but some are more critical:

TIER 1 - NON-NEGOTIABLE (Red Light Indicators)

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1. VOLUME >200% Average (30% weight) 🔴
   Purpose: Proves institutions are actually trading
   Prevents: Fading retail noise that reverses
   Without it: Trading low liquidity = slippage, fake moves
   Example: Volume 180K vs 533K required = SKIP TRADE

2. TIME WINDOW (25% weight) 🔴
   Purpose: Probability shifts dramatically by time
   Prevents: 50/50 coin flip trades during lunch
   Without it: Win rate drops from 85% to 50%
   Example: Perfect setup at 12:30 PM = SKIP (wrong window)

TIER 2 - VALUE FILTER (Yellow Light)

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3. VXX/VIX RATIO >1.45 (25% weight) 🟡
   Purpose: Tells you if VXX is mathematically overpriced
   Prevents: Fading VXX when already cheap (limited downside)
   Without it: Don't know WHEN to fade
   Scaling:
   - <1.35 = Hard NO (VXX too cheap)
   - 1.35-1.45 = Skip or tiny size
   - 1.45-1.55 = Standard position
   - 1.55-1.60 = Full position
   - >1.60 = Maximum aggression
   Example: Ratio 1.38 with perfect setup = SKIP

TIER 3 - TECHNICAL CONFIRMATION (Green Lights)

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4. PRICE AT KEY LEVEL (15% weight) 🟢
   Purpose: Identifies WHERE the turn happens
   Prevents: Entering mid-range with no edge
   Without it: Random entries with no structure
   Examples: 50 SMA, VWAP, round numbers, prior highs

5. CANDLESTICK PATTERN (5% weight) 🟢
   Purpose: Shows rejection is actually happening NOW

Prevents: Early entry before reversal confirmed

Without it: Wrong timing, get stopped out

Examples: Shooting star, doji, engulfing

THE CRITICAL INSIGHT: ALL FIVE REQUIRED

Why Not Just Use Top 3 (80% of weight)?

Scenario: Volume + Time + Ratio Perfect (80% weight)
└─ But no key level or pattern
  └─ WHERE do you enter?
    └─ Trying to catch falling knife
      └─ Result: Stopped out despite "good probability"

It's like flying:
├─ Engines (30%) = Critical
├─ Navigation (25%) = Critical
├─ Fuel (25%) = Critical
├─ Altitude (15%) = Critical
└─ Landing gear (5%) = Critical

Can't skip "just" landing gear because it's "only 5%"

You need ALL systems to land safely

HOW INDICATORS WORK TOGETHER (The Dependency Chain)

Each Step Validates the Next:

STEP 1: VOLUME >200% + TIME WINDOW
↓
Result: "Something real is happening at optimal time"
↓
STEP 2: + RATIO >1.45
↓
Result: "VXX is overpriced for this move"
↓
STEP 3: + KEY LEVEL
↓
Result: "This is where it should turn"
↓
STEP 4: + PATTERN
↓
Result: "Turn is actually happening NOW"
↓
= ENTER TRADE

Without each step, the next becomes unreliable:

- Volume + Time BUT no ratio = High activity, but is VXX expensive enough?
- Volume + Time + Ratio BUT no level = Conviction but where to enter?
- All above BUT no pattern = Entry point but no timing confirmation

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REAL SCENARIOS - WHAT BEATS WHAT

Scenario A: Perfect Ratio, No Volume

Setup Check:
├─ Ratio: 1.65 (premium fade zone) ✅
├─ Time: 3:15 PM (power hour) ✅
├─ Level: At 50 SMA resistance ✅
├─ Pattern: Shooting star forming ✅
└─ Volume: 180K (only 67% of avg) ❌

DECISION: SKIP TRADE
Reason: No institutional confirmation
Likely outcome: False signal, reverses quickly
Lesson: Volume is non-negotiable

Scenario B: Perfect Volume, Weak Ratio

Setup Check:
├─ Volume: 650K (244% average) ✅
├─ Time: 3:15 PM (power hour) ✅
├─ Level: At 50 SMA resistance ✅
├─ Pattern: Shooting star forming ✅
└─ Ratio: 1.38 (below 1.45 minimum) ❌

DECISION: SKIP TRADE
Reason: VXX not expensive enough to fade
Likely outcome: Limited downside, poor risk/reward
Lesson: Ratio is your value filter

Scenario C: All Perfect, Wrong Time

Setup Check:
├─ Ratio: 1.58 (strong fade) ✅
├─ Volume: 620K (232% average) ✅
├─ Level: At 50 SMA resistance ✅
├─ Pattern: Shooting star forming ✅
└─ Time: 12:45 PM (lunch period) ❌

DECISION: SKIP TRADE
Reason: Outside high-probability windows
Likely outcome: 50/50 coin flip despite perfect setup
Lesson: Time windows shift probability dramatically

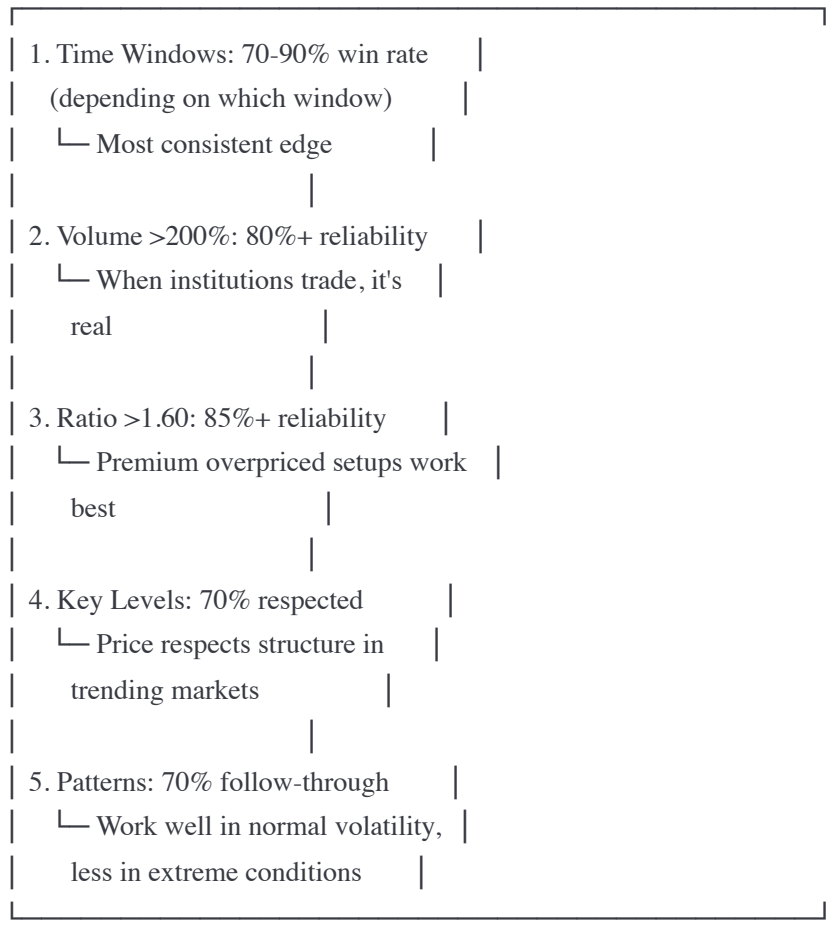
All three scenarios = SKIP with one criteria missing This proves: ALL required, no hierarchy saves you

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WHICH INDICATORS ARE MOST RELIABLE?

Historical Reliability When Present:

Ranked by Success Rate:



CRITICAL NOTE: These stats are when ALL criteria met
Individually, each is <60% reliable
Together, they create 80-90% edge

THE PILOT CHECKLIST APPROACH

Why You Can't Skip Criteria:

PRE-FLIGHT CHECKLIST:

```
┌──────────────────────────────────────────┐
│ [ ] Engines working (Volume)    │ ← 30% critical
│ [ ] Fuel sufficient (Ratio)     │ ← 25% critical
│ [ ] Navigation active (Time)    │ ← 25% critical
│ [ ] Altitude correct (Level)    │ ← 15% critical
│ [ ] Flaps deployed (Pattern)    │ ← 5% critical
│                                 │
│ ALL GREEN = Cleared for takeoff │
│ ONE RED = Abort mission         │
└──────────────────────────────────────────┘
```

You don't rank "engine vs navigation"
You need BOTH working TOGETHER
One fails = abort mission

Same with trading criteria:
Not about weight hierarchy
About SYNERGY and COMPLETENESS

WHAT EACH INDICATOR PREVENTS (The Traps)

Understanding what you're avoiding helps maintain discipline:

WITHOUT VOLUME CONFIRMATION:
├─ Trap: Trading retail noise
├─ Result: Fake moves, quick reversals
└─ Example: VXX "spike" on 50K volume reverses in 2 minutes

WITHOUT TIME WINDOW:
├─ Trap: Random probability (50/50 coin flip)
├─ Result: Inconsistent results despite "good setups"
└─ Example: Perfect pattern at lunch = 50% win rate

WITHOUT RATIO FILTER:
├─ Trap: Fading VXX when already cheap
├─ Result: Limited downside = stopped out
└─ Example: Fade at ratio 1.32, VXX only drops 2%

WITHOUT KEY LEVEL:
├─ Trap: Entering mid-range
├─ Result: No structure, no edge
└─ Example: Fade at $28.15 (between $28 and $29)

WITHOUT PATTERN:
├─ Trap: Early entry before reversal
├─ Result: Premature, gets stopped out before working
└─ Example: Enter at first push to resistance, VXX spikes higher first

---

THE DEPENDENCY TRUTH

Indicators Work in Sequence, Not Isolation:

QUESTION: "What's most important indicator?"
ANSWER: Depends on what you mean...

For PROVING MOVE IS REAL:
└─ Volume (can't fake institutional participation)

For KNOWING WHEN TO FADE:
└─ Ratio (tells you VXX is overpriced)

For MAXIMIZING PROBABILITY:
└─ Time window (shifts odds from 50% to 85%)

For IDENTIFYING ENTRY:
└─ Key level (tells you where)

For TIMING ENTRY:
└─ Pattern (tells you when)

But in practice?
└─ Need ALL FIVE working together
  └─ Like asking "what's most important:
  heart or lungs?"
  └─ You need both to live

ZERO TOLERANCE vs. FLEXIBILITY

Which Criteria Can You Bend? (Honest Assessment)

ZERO TOLERANCE (Never Break):

```
┌──────────────────────────────────────────┐
│ 1. VOLUME >200%                          │
│    └─ No exceptions, ever                │
│    └─ This is your proof                 │
│                                          │
│ 2. TIME WINDOW                           │
│    └─ No exceptions, ever                │
│    └─ This is when probability           │
│      shifts                              │
└──────────────────────────────────────────┘
```

MINIMAL FLEXIBILITY (Rarely):

```
┌──────────────────────────────────────────┐
│ 3. RATIO >1.45                           │
│    Hard floor: 1.42 too low              │
│    Acceptable: 1.45-1.47                 │
│    Prefer: >1.50                         │
│    Ideal: >1.60                          │
└──────────────────────────────────────────┘
```

SOME FLEXIBILITY (Context):

```
┌──────────────────────────────────────────┐
│ 4. KEY LEVEL                             │
│    Ideal: Exact 50 SMA touch             │
│    Acceptable: Within $0.20              │
│    Skip: Mid-range                       │
│                                          │
│ 5. PATTERN                               │
│    Ideal: Perfect shooting star          │
│    Acceptable: Rejection wick            │
│    Skip: No pattern at all               │
└──────────────────────────────────────────┘
```

THE SYNERGY FORMULA

How 5 Weak Signals Create 1 Strong Signal:

Individual Reliability:
├─ Volume spike alone: 55% reliable
├─ Ratio >1.50 alone: 60% reliable
├─ Time window alone: 65% reliable
├─ Key level alone: 50% reliable
└─ Pattern alone: 55% reliable

Combined (All 5):
└─ All criteria met: 85%+ reliable ✅

This is NOT additive (5 × 60% ≠ 300%)
This is MULTIPLICATIVE validation:
└─ Each confirms the other
  └─ Probability compounds
    └─ Edge emerges from synergy

PRACTICAL APPLICATION AT 3:10 PM

How to Use This Knowledge in Real-Time:

Step 1: Check in Priority Order

1. Is it 3:10-3:25 PM? (Time)
  └─ NO → Stop, skip trade
  └─ YES → Continue

2. Is volume >533K? (Volume)
  └─ NO → Stop, skip trade
  └─ YES → Continue

3. Is ratio >1.45? (Ratio)
  └─ NO → Stop, skip trade
  └─ YES → Continue, note if >1.55 for sizing

4. Is VXX at resistance? (Level)
  └─ NO → Wait or skip
  └─ YES → Continue

5. Is pattern forming? (Pattern)
  └─ NO → Wait 1 candle
  └─ YES → ENTER TRADE

Get to Step 3 and ratio is 1.42?

- Stop checking immediately
- Skip trade completely
- Don't care if #4 and #5 are perfect
- No second-guessing

All 5 green?

- Enter immediately
- Don't overthink
- Trust the checklist
- Execute mechanically

WHY THIS HIERARCHY MATTERS

Benefits of Understanding Weight & Purpose:

CLARITY:
├─ Know WHY you're checking each thing
├─ Not just THAT you should check it
└─ Builds confidence in system

DISCIPLINE:
├─ Understand what each prevents
├─ Less tempted to skip criteria
└─ "This prevents fading cheap VXX" = keep the rule

EXECUTION SPEED:
├─ Check in priority order (time → volume → ratio)
├─ Abort early if non-negotiables fail
└─ Don't waste time if red light already showing

POSITION SIZING:
├─ Ratio 1.48 vs 1.62 = different conviction
├─ Adjust size based on strength
└─ Maximum aggression when all align strongly

FINAL TRUTH ABOUT INDICATORS

```
THE QUESTION: "Which indicator is most important?"

THE ANSWER: "Wrong question."

BETTER QUESTION: "Why do I need all 5?"

THE TRUTH:
├─ Volume proves it's real (institutions trading)
├─ Ratio proves it's expensive (worth fading)
├─ Time proves probability is high (optimal window)
├─ Level proves there's a turn point (structure)
└─ Pattern proves turn is happening (timing)

Remove any one:
└─ Missing proof in chain
   └─ Edge disappears
      └─ Back to 50/50 gambling

Keep all five:
└─ Complete proof chain
   └─ 85%+ edge emerges
      └─ Systematic profits compound
```

Remember: You're a pilot, not a gambler

Pre-flight checklist must be 100% green

No hierarchy saves you from missing critical system

Trust the complete checklist. Execute mechanically.

VXX/VIX Ratio (Primary)

```thinkscript
```

```
declare lower;

def vix = close("VIX");
def vxx = close("VXX");
def ratioValue = vxx / vix;

plot RatioLine = ratioValue;
plot HighThreshold = 1.60;
plot LowThreshold = 1.35;

RatioLine.SetDefaultColor(Color.CYAN);
HighThreshold.SetDefaultColor(Color.GREEN);
LowThreshold.SetDefaultColor(Color.RED);

AddCloud(RatioLine, HighThreshold, Color.DARK_GREEN, Color.CURRENT);
```

Save as: VXX_VIX_Ratio

Optional: VXX/VIX Ratio with Labels (Advanced)

```
thinkscript
```

```
declare lower;

def vix = close("VIX");
def vxx = close("VXX");
def ratioValue = vxx / vix;

plot RatioLine = ratioValue;
plot HighThreshold = 1.60;
plot MidThreshold = 1.475;
plot LowThreshold = 1.35;

RatioLine.SetDefaultColor(Color.CYAN);
RatioLine.SetLineWeight(2);
HighThreshold.SetDefaultColor(Color.GREEN);
MidThreshold.SetDefaultColor(Color.GRAY);
LowThreshold.SetDefaultColor(Color.RED);

# Color coding
RatioLine.AssignValueColor(
    if ratioValue > 1.60 then Color.GREEN
    else if ratioValue > 1.55 then Color.YELLOW
    else if ratioValue > 1.45 then Color.CYAN
    else if ratioValue > 1.35 then Color.ORANGE
    else Color.RED
);

AddCloud(RatioLine, HighThreshold, Color.DARK_GREEN, Color.CURRENT);
AddCloud(LowThreshold, RatioLine, Color.DARK_RED, Color.CURRENT);

# Labels
AddLabel(yes, "Ratio: " + Round(ratioValue, 2),
    if ratioValue > 1.60 then Color.GREEN
    else if ratioValue > 1.45 then Color.YELLOW
    else Color.RED
);

AddLabel(yes,
    if ratioValue > 1.60 then "PREMIUM FADE"
    else if ratioValue > 1.55 then "STRONG FADE"
    else if ratioValue > 1.45 then "NORMAL FADE"
    else if ratioValue > 1.35 then "WEAK FADE"
    else "NO FADE",
    if ratioValue > 1.60 then Color.GREEN
```

```
    else if ratioValue > 1.45 then Color.YELLOW
    else Color.RED
);
```

Save as: VXX_VIX_Ratio_Advanced

This version adds:

- Color-coded line (green when premium fade, red when no fade)
- Labels showing current ratio and trade recommendation
- Visual zones for all ratio tiers

Optional: Volume Confirmation Indicator

```
thinkscript

declare lower;

def volAvg = Average(volume, 30);
def volPercent = (volume / volAvg) * 100;

plot VolPercent = volPercent;
plot Threshold200 = 200;
plot Threshold300 = 300;

VolPercent.SetDefaultColor(Color.CYAN);
VolPercent.SetLineWeight(2);
Threshold200.SetDefaultColor(Color.YELLOW);
Threshold300.SetDefaultColor(Color.GREEN);

VolPercent.AssignValueColor(
    if volPercent >= 300 then Color.GREEN
    else if volPercent >= 200 then Color.YELLOW
    else Color.RED
);

AddLabel(yes, "Vol: " + Round(volPercent, 0) + "%",
    if volPercent >= 200 then Color.GREEN else Color.RED
);

AddCloud(VolPercent, Threshold300, Color.DARK_GREEN, Color.CURRENT);
```

Save as: Volume_Percent_Tracker

Shows volume as % of 30-period average with color coding

TROUBLESHOOTING

If VXX/VIX Ratio doesn't appear:

1. Check you're on VXX chart (not VIX)
2. Verify it's applied to "Lower Panel"
3. Check for typos in code
4. Restart ThinkOrSwim if needed

If alerts don't trigger:

1. Verify alerts are enabled (bell icon should be on)
2. Check computer volume is up
3. Test with temporary price alert
4. Confirm "Once per bar" setting

If volume bars wrong colors:

1. Edit Volume study
2. Verify "Show Volume Average" is checked
3. Confirm colors: Green (above), Red (below)
4. Check Average Length = 30

If chart loads slowly:

1. Reduce chart history to 5-10 days
2. Close unused tabs
3. Restart ThinkOrSwim daily

PERFORMANCE OPTIMIZATION

Best Practices:

- **Load only VXX chart** (add VIX as separate tab if needed)

- **Limit to 5-10 days history** (enough for pattern recognition)
- **Close platform overnight** (fresh start daily)
- **Test alerts weekly** (set temporary test alerts)

Chart Settings for Speed:

1. Setup → Application Settings → Charts
2. **Days to load**: 10 days
3. **Default aggregation**: 5 min
4. **Disable**: Real-time news (use separate browser)

WHY THIS SETUP WORKS

The Complete Picture:

1. **VXX/VIX Ratio** = Tells you WHEN to fade (value filter)
2. **Volume** = Tells you IF institutions agree (confirmation)
3. **Moving Averages** = Tells you WHERE resistance is (levels)
4. **VWAP** = Tells you institutional bias (direction)
5. **Time Alerts** = Tells you optimal entry windows (timing)

Together = Professional trading system

The Missing Piece (VXX/VIX Ratio):

Before: You knew VXX spiked, but not if it was "expensive enough" to fade **After**: Ratio >1.60 = mathematically overpriced = high-probability fade

This is the difference between:

- "VXX is up 5%, should I fade it?" (guess)
- "VXX is up 5% AND ratio is 1.63 = premium fade setup" (systematic)

FINAL CHECKLIST

Before going live, verify:

Indicators Loaded:

- [ ] 20 SMA (yellow line visible)
- [ ] 50 SMA (orange line visible)
- [ ] VWAP (cyan line visible)
- [ ] Volume with average (green/red bars)
- [ ] Implied Volatility (lower panel)
- [ ] VXX/VIX Ratio (lower panel with green/red thresholds)

Alerts Configured:

- [ ] 9:45 AM (morning window)
- [ ] 3:05 PM (power hour)
- [ ] 3:40 PM (institutional flow)
- [ ] VXX 3% spike
- [ ] Volume surge
- [ ] VIX >25 warning

Workspace Saved:

- [ ] Named "VXX_Trading_System_2026"
- [ ] Loads all indicators automatically
- [ ] Options chain accessible

Chart Readable:

- [ ] Can clearly see ratio value
- [ ] Volume % visible in label
- [ ] Price vs moving averages obvious

DAILY PRE-FLIGHT CHECK (30 seconds)

Every morning:

1. Load workspace ✅
2. Check all 6 indicators visible ✅
3. Check ratio is calculating (shows number) ✅
4. Set test alert (VXX >1%) to verify alerts work ✅
5. Review overnight news ✅

If any indicator missing = fix before trading

THE SYSTEM IN ACTION

Example: Premium Fade Setup

9:55 AM - Morning Window

Main Chart:

- VXX at $29.50 (gapped up overnight)
- Sitting at 50 SMA (orange line resistance)
- Shooting star forming

Volume Panel:

- Current bar showing green
- Label: "5,807 Vol: 266,837" = ERROR - only 2%
- WAIT - volume not confirmed yet

10:05 AM - Volume Surge

Volume Panel:

- NOW showing: "620,000 Vol: 266,837" = 232% ✅
- Green bars stacking

VXX/VIX Ratio Panel:

- Ratio showing 1.58 (approaching green line)
- Still in normal fade zone but strong

Action: Enter puts, standard position ($300-400)

10:10 AM - Ratio Spike

VXX/VIX Ratio Panel:

- VXX pushed to $30.20
- Ratio now 1.64 (ABOVE green line)
- Premium fade territory

Action: If still in position, hold. If missed entry, this confirms setup was correct.

10:45 AM - Fade Complete

Main Chart:

- VXX dropped to $28.80
- Pattern worked
- Exit with 40% gain

The ratio confirmed this was worth fading at 1.58, became obvious at 1.64

CONCLUSION

This setup gives you:

- **Systematic entries** (no guessing)
- **Value confirmation** (ratio filter)
- **Institutional validation** (volume)
- **Optimal timing** (alerts)
- **Professional execution** (all info on one screen)

The VXX/VIX ratio is your edge. Everything else confirms it.

Setup once. Trade systematically. Remove emotion.

APPENDIX: QUICK RATIO REFERENCE

Memorize These Numbers:

- **>1.60** = Premium fade (max position)
- **1.55-1.60** = Strong fade (full position)
- **1.45-1.55** = Normal fade (standard position)
- **1.35-1.45** = Weak fade (half size or skip)
- **<1.35** = No fade (skip puts)

Historical Context:

- **Normal market**: 1.45-1.50
- **Fear spike**: 1.60-1.75
- **Extreme panic**: 1.75-2.00 (rare, max opportunity)
- **Post-spike decay**: 1.35-1.45 (recovery phase)

Your chart shows this in real-time. Trust the numbers.

Save this document. Your complete ThinkOrSwim setup is here.

If platform crashes, you can rebuild in 15 minutes.